

Основни тъждества

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\operatorname{tg} \alpha = \frac{\sin \alpha}{\cos \alpha}$$

$$1 + \operatorname{tg}^2 \alpha = \frac{1}{\cos^2 \alpha}$$

$$\operatorname{tg} \alpha \cdot \operatorname{cotg} \alpha = 1$$

$$\operatorname{cotg} \alpha = \frac{\cos \alpha}{\sin \alpha}$$

$$1 + \operatorname{cotg}^2 \alpha = \frac{1}{\sin^2 \alpha}$$

Сбор и разлика на 2 ъгъла

$$\sin(\alpha \pm \beta) = \sin \alpha \cdot \cos \beta \pm \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cdot \cos \beta \mp \sin \alpha \cdot \sin \beta$$

$$\operatorname{tg}(\alpha \pm \beta) = \frac{\operatorname{tg} \alpha \pm \operatorname{tg} \beta}{1 \mp \operatorname{tg} \alpha \cdot \operatorname{tg} \beta}$$

$$\operatorname{cotg}(\alpha \pm \beta) = \frac{\operatorname{cotg} \alpha \cdot \operatorname{cotg} \beta \mp 1}{\operatorname{cotg} \beta \pm \operatorname{cotg} \alpha}$$

Връзки между тригонометричните функции

	$\sin \alpha$	$\cos \alpha$	$\operatorname{tg} \alpha$	$\operatorname{cotg} \alpha$
$\sin \alpha$	$\sin \alpha$	$\pm \sqrt{1 - \cos^2 \alpha}$	$\frac{\operatorname{tg} \alpha}{\pm \sqrt{1 + \operatorname{tg}^2 \alpha}}$	$\frac{1}{\pm \sqrt{1 + \operatorname{cotg}^2 \alpha}}$
$\cos \alpha$	$\pm \sqrt{1 - \sin^2 \alpha}$	$\cos \alpha$	$\frac{1}{\pm \sqrt{1 + \operatorname{tg}^2 \alpha}}$	$\frac{\operatorname{cotg} \alpha}{\pm \sqrt{1 + \operatorname{cotg}^2 \alpha}}$
$\operatorname{tg} \alpha$	$\frac{\sin \alpha}{\pm \sqrt{1 - \sin^2 \alpha}}$	$\frac{\pm \sqrt{1 - \cos^2 \alpha}}{\cos^2 \alpha}$	$\operatorname{tg} \alpha$	$\frac{1}{\operatorname{cotg} \alpha}$
$\operatorname{cotg} \alpha$	$\frac{\pm \sqrt{1 - \sin^2 \alpha}}{\sin^2 \alpha}$	$\frac{\cos \alpha}{\pm \sqrt{1 - \cos^2 \alpha}}$	$\frac{1}{\operatorname{tg} \alpha}$	$\operatorname{cotg} \alpha$

Двойни и половинки ъгли

$$\sin 2\alpha = 2\sin \alpha \cdot \cos \alpha \qquad \cos \alpha \pm \sin \alpha = \sqrt{1 \pm \sin 2\alpha}$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = \frac{1}{2} - 2\sin^2 \alpha = 2\cos^2 \alpha - 1$$

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$\cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}$$

$$\sin \alpha = \frac{2\operatorname{tg} \frac{\alpha}{2}}{1 + \operatorname{tg}^2 \frac{\alpha}{2}}$$

$$\cos \alpha = \frac{1 - \operatorname{tg}^2 \frac{\alpha}{2}}{1 + \operatorname{tg}^2 \frac{\alpha}{2}}$$

Понижаване на степен

$$2 \sin^2 \alpha = 1 - \cos 2\alpha$$

$$2 \cos^2 \alpha = 1 + \cos 2\alpha$$

Други

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

Тангенсова Теорема

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$\frac{a+b}{a-b} = \frac{\operatorname{tg} \frac{a+b}{2}}{\operatorname{tg} \frac{a-b}{2}}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2} = 2 \sin \frac{\alpha + \beta}{2} \sin \frac{\beta - \alpha}{2}$$